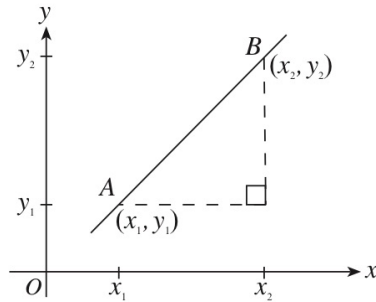




Gradient

1. The gradient of the line joining any two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1} \text{ or } \frac{y_1 - y_2}{x_1 - x_2}.$$
2. Parallel lines have the same gradient.

Distance

3. The distance between any two points $A(x_1, y_1)$ and $B(x_2, y_2)$ is given by

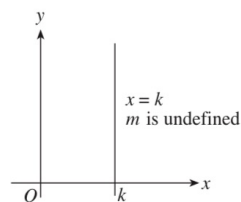
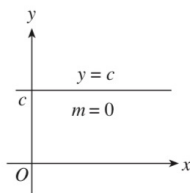
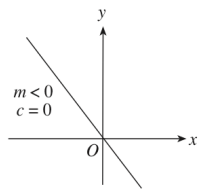
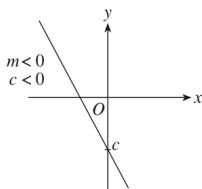
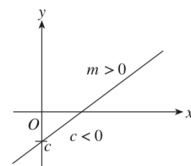
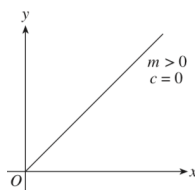
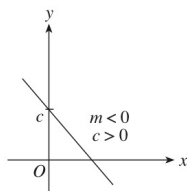
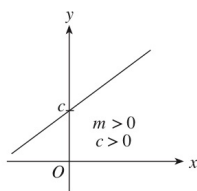
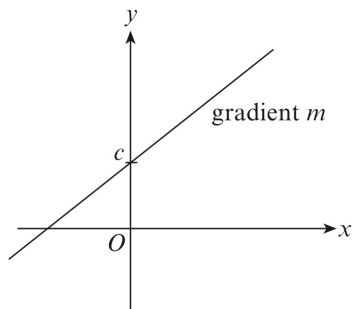
$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

The gradient of the line joining the points $A(x, 9)$ and $B(2, 8)$ is $\frac{1}{2}$.

- (a) Find the value of x .
- (b) Find the length of AB .

Equation of a Straight Line

4. The equation of the straight line with gradient m and y-intercept c is $y = mx + c$.



E.g 2

A line passes through the points $A(6, 2)$ and $B(5, 5)$. Find the equation of the line.

5. The equation of a horizontal line is of the form $y = c$.
6. The equation of a vertical line is of the form $x = k$.

11. The points A , B and C are $(8, 7)$, $(11, 3)$ and $(3, -3)$ respectively.
- (a) Find the equation of the line parallel to AB and passing through C .
 - (b) Show that AB is perpendicular to BC .
 - (c) Calculate the area of triangle ABC .

Q2. P is the point $(-4, 10)$ and Q is the point $(5, -8)$.

(a) Find the length of line PQ .

Answer [2]

(b) Find the equation of line PQ .

Answer [2]

(c) R is the point where the line PQ cuts the x -axis.

Find the coordinates of R .

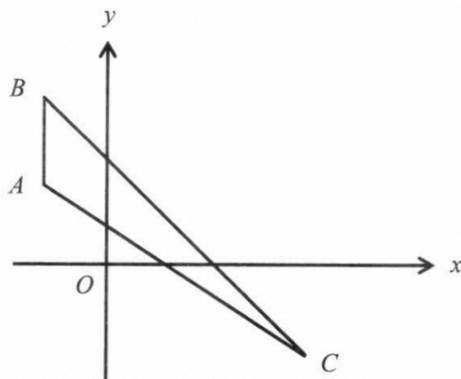
Answer R (..... ,)

- (d) The equation of line k is $3x + 2y = 7$.

Find the coordinates of the point of intersection of the line PQ and line k .

Answer (.....,) [3]

Q3.



The diagram shows a triangle ABC with vertices $A(-2, 4)$ and $C(10, -4)$.
 AB is parallel to the y -axis.

- (a) Calculate the length of the line segment AC .

Answer units [2]

- (b) Given that the gradient of the line $BC = -1$.

Find

- (i) the coordinates of B ,

Answer B (.....,) [2] EM Notes

- (ii) the equation of the line BC .

Answer [2]

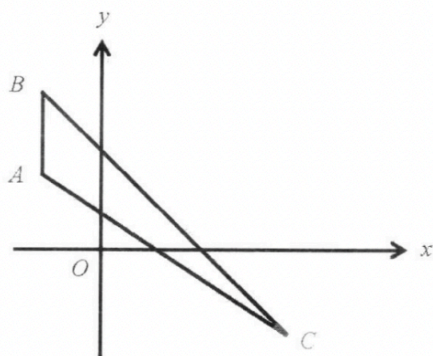
- (c) Calculate the area of the triangle ABC .

Answer units^2 [2]

- (d) Find the coordinates of point D , given that $ABCD$ is a parallelogram.

Answer D (.....,) [1]

Answers



The diagram shows a triangle ABC with vertices $A(-2, 4)$ and $C(10, -4)$.
 AB is parallel to the y -axis.

- (a) Calculate the length of the line segment AC .

$$\begin{aligned} AC &= \sqrt{(-2-10)^2 + [4-(-4)]^2} && \text{[M1]} \\ &= \sqrt{208} \\ &= 14.4 \text{ units (3 s. f.)} && \text{[A1]} \end{aligned}$$

14.4
 Answer units [2]

- (b) Given that the gradient of the line $BC = -1$.

Find

- (i) the coordinates of B ,

Let the coordinates of B be $(-2, b)$.

$$\begin{aligned} \frac{b-(-4)}{-2-10} &= -1 && \text{[M1]} \\ b+4 &= 12 \\ b &= 8 \\ B &= (-2, 8) && \text{[A1]} \end{aligned}$$

-2 8
 Answer B (.....,) [2]

(ii) the equation of the line BC .

$$y = mx + c$$

$$-4 = (-1)(10) + c \quad [\text{M1}]$$

$$c = 6$$

$$\text{Equation of } BC \text{ is } y = -x + 6 \quad [\text{A1}] \text{ reject } y = -1x + 6$$

$$y = -x + 6$$

Answer [2]

(c) Calculate the area of the triangle ABC .

$$\text{Area of triangle} = \frac{1}{2} \times 4 \times 12 \quad [\text{M1}]$$

$$= 24 \text{ units}^2 \quad [\text{A1}]$$

$$24$$

Answer units² [2]

(d) Find the coordinates of point D , given that $ABCD$ is a parallelogram.

$$10$$

$$-8$$

Answer D (.....,) [1]